

AN ENERGY BOUND FOR AVOIDANCE INDICATORS, AND FOURIER TAILS OF DNFs OVER PRODUCT SPACES

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ABSTRACT. Let F be the indicator that none of a finite family of cylinder events occurs, on an arbitrary finite product probability space, and let $F = \sum_U F^{=U}$ be its Efron–Stein decomposition. We prove that

$$\sum_U \left(\prod_{v \in U} \lambda_v \right) \|F^{=U}\|_2^2 \leq 1$$

for all weights $\lambda_v \geq 1$ such that $\prod_{v \in \text{supp } E} \lambda_v \leq 2$ for every event E . There is no dependence on the number of events, the alphabet sizes or the measure. In particular every width- k DNF $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$ satisfies $\sum_{|S| > t} \hat{g}(S)^2 \leq 4 \cdot 2^{-(t+1)/k}$, under every product measure; so g is ε -concentrated up to degree $k \log_2(4/\varepsilon)$. The proof does not use a switching lemma. It bounds the Efron–Stein energies by signed cover counts of the events that hold at a random point, writes the resulting hypergraph functional as the second moment of an alternating union sum under a random deletion, and bounds that sum by deletion–contraction. Over general product spaces the base $2^{-1/k}$ is sharp, and the threshold 2 is necessary. We also give a version with weights that grow along a filtration of each coordinate, and we explain the application that motivated it: the junta term in a lower bound for the least “multiplier modulus” of the Erdős–Straus equation.

1. INTRODUCTION

Let $\Omega = \prod_{v \in V_0} \Omega_v$ be a finite product of finite sets with a product probability measure $\pi = \bigotimes_v \pi_v$. A *cylinder event* fixes the values of the coordinates in a set $\text{supp } E \subseteq V_0$, its *support*. For a finite family $\mathcal{E}$ of cylinder events let

$$F = F_{\mathcal{E}} = \prod_{E \in \mathcal{E}} (1 - \mathbf{1}_E)$$

be its *avoidance indicator*, and let $F = \sum_{U \subseteq V_0} F^{=U}$ be the Efron–Stein (Hoeffding) decomposition of F in $L^2(\pi)$ [4, 6], [10, §8.3]. For the uniform measure on $\{\pm 1\}^n$ this is the Fourier expansion, $F^{=U} = \hat{F}(U)\chi_U$. We study the weighted energy

$$G_F(\lambda) = \sum_{U \subseteq V_0} \lambda^U \|F^{=U}\|^2, \quad \lambda^U = \prod_{v \in U} \lambda_v, \quad \lambda_v \geq 1.$$

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For constant weights $\lambda_v = \rho^2$ it is $\|T_\rho F\|^2$, where T_ρ multiplies the component $F^=U$ by $\rho^{|U|}$; so we are asking for the largest noise operator *with* $\rho > 1$ that still maps every avoidance indicator (of a suitable family) into the unit ball of L^2 . (It is not a contraction in L^2 : $\|T_\rho F\| > \|F\|$ for nonconstant F .)

Theorem 1.1 (Energy bound). *Let $\lambda_v \geq 1$ ($v \in V_0$), and suppose that $\lambda^{\text{supp } E} \leq 2$ for every $E \in \mathcal{E}$. Then*

$$G_F(\lambda) = \sum_U \lambda^U \|F^=U\|^2 \leq 1.$$

The hypothesis is on single events only. There is no condition on the number of events, on their masses or overlaps, on the alphabet sizes $|\Omega_v|$ or on the measure. The constant 2 cannot be raised (Proposition 2.3). Since $\sum_{U: \lambda^U > e^\tau} \|F^=U\|^2 \leq e^{-\tau} G_F(\lambda)$, the theorem is a family of tail bounds, one for each choice of weights. Constant weights give the following.

Corollary 1.2 (Energy tail). *If every event of $\mathcal{E}$ has $|\text{supp } E| \leq k$ ($k \geq 1$), then for $f \in \{F, 1 - F\}$ and every integer $t \geq 0$,*

$$\text{En}(f; t) := \sum_{|U| > t} \|f^=U\|^2 \leq 2^{-(t+1)/k} \mathbb{P}(F = 0)(2 - \mathbb{P}(F = 0)) \leq 2^{-(t+1)/k}.$$

A width- k DNF is $g = 2F - 1$ for the family of its terms (with the convention $-1 = \text{True}$), and the corollary becomes a statement about its Fourier tail.

Corollary 1.3 (Fourier tails of DNFs). *Let $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$ be computed by a DNF of width k , and let π be any product measure on $\{\pm 1\}^n$ (Efron–Stein components in $L^2(\pi)$; Fourier coefficients if π is uniform). Then for every integer $t \geq 0$*

$$W^{>t}[g] := \sum_{|U| > t} \|g^=U\|^2 \leq 4p(2-p) \cdot 2^{-(t+1)/k} \leq 4 \cdot 2^{-(t+1)/k},$$

where $p = \mathbb{P}[g = -1]$. So g is ε -concentrated up to degree $k \log_2(4/\varepsilon)$. Moreover $\sum_U 2^{|U|/k} \|g^=U\|^2 \leq 1 + 4p$, and the total influence $\mathbf{I}[g] := \sum_U |U| \|g^=U\|^2$ is at most $(4k/\ln 2)p$. The same holds for q -ary DNFs, with literals $x_v \in S_v$, under any product measure on $\prod_v [q_v]$.

Comparison. The classical bound goes back to Linial, Mansour and Nisan [8], via Håstad’s switching lemma [5]: width- w DNFs are ε -concentrated up to degree $O(w \log(1/\varepsilon))$. In the form of O’Donnell’s book [10, §4.4] it reads $W^{\geq t}[g] \leq 2 \cdot 2^{-t/(20w)}$ (we quote the constant 20 from memory and could not re-check it). It has been used in this form, with an unspecified constant C , in later work on DNFs, for instance by Mansour [9] (from memory) and by Lecomte and Tan [7, Fact 6]. Corollary 1.3 has the constant 1 in the exponent and holds for arbitrary product measures and alphabets. Over general product spaces the base $2^{-1/k}$ is sharp: no bound $C\rho^{-t/k}$ with $\rho > 2$ holds, and at most a factor of order $\sqrt{t/k}$ can be gained (Proposition 6.1). On the uniform cube we only know that the true rate lies between $3^{-t/k}$

and $2^{-t/k}$ (Example 6.2). The influence bound is *not* a gain: an elementary certificate argument gives $\mathbf{I}[g] \leq 2kp$ on the uniform cube and $\sum_v \|L_v g\|^2 \leq 4kp$ under any product measure (Remark 5.1), so our $(4k/\ln 2)p$ is weaker by the factor $2/\ln 2$, resp. $1/\ln 2$. Sharper bounds without the factor p are known (from memory: $\mathbf{I} \leq 2w$ is usually attributed to Boppana [3], and we recall that Amano [1] improved it to $\mathbf{I} \leq w$). We state the influence bound only because it falls out.

The nearest prior art we know is the paper of Lecomte and Tan [7]. They bound $|\hat{g}(S)|$ by $2^{|S|}$ times the probability that S is covered by the terms satisfied at a random point. This is the device of our Lemma 3.1, but there are three differences: our cover counts are *signed*, and vanish as soon as a satisfied term misses S ; we work in L^2 with all levels and weights at once; and our degree concentration does not go through the switching lemma. The two combinatorial steps of the proof, a polarization identity (Lemma 4.1) and a deletion–contraction bound (Lemma 4.2), have no counterpart there.

What we checked. We have no internet access while writing this draft, and our literature search is partial. Earlier, [7] and Tal [13] were checked online; in [13] we found no explicit-constant tail bound for width- w DNFs. Everything else above is quoted from memory and is marked as such. We did not search systematically through the switching-lemma literature (sharp variants of Håstad’s lemma, Rossman’s entropy proof), the literature on Fourier growth, or work on noise operators with $\rho > 1$. We believe Theorem 1.1 and the constant 1 in Corollary 1.3 are new, but we make no priority claim.

Proof outline. Write $G_F = \sum_V \mu^V \|L_V F\|^2$ with $\mu_v = \lambda_v - 1$ and $L_V = \prod_{v \in V} (I - \mathbb{E}_v)$. Section 3 bounds $\|L_V F\|^2$ by the mean square of a signed count $N_{\mathcal{H}(x)}(V)$ of covers of V by the events that hold at x , and so reduces Theorem 1.1 to the hypergraph inequality $Q_\mu(\mathcal{H}) = \sum_V \mu^V N_{\mathcal{H}}(V)^2 \leq 1$. Section 4 proves it: $Q_\mu(\mathcal{H})$ is the second moment of the alternating sum $\Theta_\lambda(\mathcal{H}_P) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \lambda^{\bigcup \mathcal{J}}$ over the edges that miss a random set P , and deletion–contraction gives $|\Theta| \leq 1$. Nothing in the proof is specific to two-point spaces or to the uniform measure.

Organisation. Section 2 sets up notation. Sections 3 and 4 prove Theorem 1.1. Section 5 proves the corollaries, Section 6 discusses sharpness, failed strengthenings and open problems, and Section 7 gives the filtration variant. Section 8 sketches the application to the Erdős–Straus equation [12]. The exact-arithmetic script `scripts/energy_note_check.py` checks the main inequalities on small cases (Section 9).

2. SETTING AND ELEMENTARY FACTS

Product spaces. Throughout, $\Omega = \prod_{v \in V_0} \Omega_v$ is a finite product of finite sets, V_0 is finite, and $\pi = \bigotimes_v \pi_v$ is a product probability measure. We write $\mathbb{P}$, $\mathbb{E}$, $\langle \cdot, \cdot \rangle$ and $\|\cdot\| = \|\cdot\|_{L^2(\pi)}$. For $W \subseteq V_0$ we write x_W for the restriction of $x \in \Omega$ to W , $\Omega_W = \prod_{v \in W} \Omega_v$ and $\pi_W = \bigotimes_{v \in W} \pi_v$. A *cylinder event* is a set $E = \{x : x_v = \sigma_v \text{ (} v \in \text{supp } E)\}$ with $\text{supp } E \subseteq V_0$ and

$\sigma \in \Omega_{\text{supp } E}$. A family $\mathcal{E}$ of cylinder events is finite, and repetitions are allowed. Its avoidance indicator is $F = \prod_{E \in \mathcal{E}} (1 - \mathbf{1}_E)$, and $h = 1 - F$ is the indicator that some event occurs. For weights $\lambda_v \geq 1$ we put $\mu_v = \lambda_v - 1 \geq 0$, $\lambda^U = \prod_{v \in U} \lambda_v$, $\mu^U = \prod_{v \in U} \mu_v$, and $w_E = \lambda^{\text{supp } E}$ (the *weight* of E).

Efron–Stein decomposition (classical; [4, 6], [10, §8.3]). For $\varphi : \Omega \rightarrow \mathbb{R}$ and $W \subseteq V_0$ let $\mathbb{E}[\varphi \mid x_W]$ be the conditional expectation given the coordinates in W , and

$$\varphi^{=U} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} \mathbb{E}[\varphi \mid x_W].$$

Then $\varphi = \sum_U \varphi^{=U}$, the components are mutually orthogonal, $\varphi^{=U}$ depends only on x_U , and $\mathbb{E}_u \varphi^{=U} = 0$ for $u \in U$, where $\mathbb{E}_u$ averages over the coordinate u (with respect to π_u). Put $L_u = I - \mathbb{E}_u$ and $L_V = \prod_{u \in V} L_u$. The L_u are commuting orthogonal projections, and $L_V \varphi = \sum_{U \supseteq V} \varphi^{=U}$. If $\mathcal{D}$ is a down-closed family of subsets of V_0 , then $\varphi^{\mathcal{D}} = \sum_{U \in \mathcal{D}} \varphi^{=U}$ is a linear combination of the functions $\mathbb{E}[\varphi \mid x_W]$, $W \in \mathcal{D}$ (by Möbius inversion), and

$$\mathbb{E}(\varphi - \varphi^{\mathcal{D}})^2 = \sum_{U \notin \mathcal{D}} \|\varphi^{=U}\|^2. \quad (1)$$

For $\mathcal{D} = \{U : |U| \leq t\}$ this is the *energy above level t* , $\text{En}(\varphi; t)$. It is the least value of $\mathbb{E}(\varphi - g)^2$ over sums g of functions that each depend on at most t coordinates. For ± 1 -valued g we also write $W^{>t}[g] = \text{En}(g; t)$, as in [10].

Definition 2.1 (Energy generating function). For weights $\lambda_v \geq 1$,

$$G_\varphi(\lambda) = \sum_{U \subseteq V_0} \lambda^U \|\varphi^{=U}\|^2.$$

Lemma 2.2 (Elementary identities). *For every φ and all weights $\lambda_v \geq 1$:*

- (a) $G_\varphi(\lambda) = \sum_{V \subseteq V_0} \mu^V \|L_V \varphi\|^2$;
- (b) $\sum_{U: \lambda^U > e^\tau} \|\varphi^{=U}\|^2 \leq e^{-\tau} G_\varphi(\lambda)$ for every real τ ;
- (c) $G_{1-\varphi} = 1 - 2\mathbb{E}\varphi + G_\varphi$; in particular $G_F \leq 1$ if and only if $G_h \leq 2\mathbb{E}h$;
- (d) if $\varphi(x) = \prod_v \varphi_v(x_v)$ is a product, then $G_\varphi(\lambda) = \prod_v G_{\varphi_v}(\lambda_v)$, where $G_{\varphi_v}(\lambda_v) = (\mathbb{E}\varphi_v)^2 + \lambda_v \text{Var } \varphi_v$.

Proof. (a) Expand $\lambda^U = \prod_{v \in U} (1 + \mu_v) = \sum_{V \subseteq U} \mu^V$ and use $\|L_V \varphi\|^2 = \sum_{U \supseteq V} \|\varphi^{=U}\|^2$. (b) is Markov's inequality. (c) $(1 - \varphi)^{=U} = -\varphi^{=U}$ for $U \neq \emptyset$ and $(1 - \varphi)^{=\emptyset} = 1 - \mathbb{E}\varphi$. (d) For a product, $\varphi^{=U} = \prod_{v \in U} (\varphi_v - \mathbb{E}\varphi_v) \prod_{v \notin U} \mathbb{E}\varphi_v$. $\square$

Proposition 2.3 (A single event; the threshold 2). *Let E fix $x_v = \sigma_v$ on $\text{supp } E$, put $p_v = \pi_v(\sigma_v)$ and $\pi_E = \mathbb{P}(E) = \prod_{v \in \text{supp } E} p_v$. Then*

$$G_{1-\mathbf{1}_E}(\lambda) = 1 - 2\pi_E + \pi_E \prod_{v \in \text{supp } E} (\lambda_v - \mu_v p_v).$$

Hence, for fixed weights with $w_E > 2$, $G_{1-\mathbf{1}_E}(\lambda) > 1$ as soon as all $p_v > 0$ are small enough. So the constant 2 in Theorem 1.1 cannot be replaced by any larger number, even for a single event.

Proof. By Lemma 2.2(d), with $\varphi_v = \mathbf{1}[x_v = \sigma_v]$, $G_{\mathbf{1}_E} = \prod_v (p_v^2 + \lambda_v p_v (1 - p_v)) = \pi_E \prod_v (\lambda_v - \mu_v p_v)$. Then use Lemma 2.2(c). The product tends to w_E as $p \rightarrow 0$. $\square$

On the uniform cube $\{\pm 1\}^n$ the single-event condition is the weaker $\prod_v (1 + \lambda_v)/2 \leq 2$. The threshold $w_E \leq 2$ is forced by large alphabets or strongly biased measures.

3. THE COVER BOUND

A *hypergraph* $\mathcal{H}$ is a finite set of subsets (*edges*) of V_0 ; the empty edge is allowed. For $V \subseteq V_0$ let

$$N_{\mathcal{H}}(V) = \sum_{\mathcal{J} \subseteq \mathcal{H}: \bigcup_{J \in \mathcal{J}} (J \cap V) = V} (-1)^{|\mathcal{J}|}, \quad Q_{\mu}(\mathcal{H}) = \sum_{V \subseteq V_0} \mu^V N_{\mathcal{H}}(V)^2.$$

So $N_{\mathcal{H}}(V)$ is a signed count of the subfamilies whose traces cover V . It depends only on the multiset of traces $J \cap V$. It does not change if an edge is replaced by several edges with the same trace: a class of $m \geq 1$ edges with equal traces contributes $\sum_{j \geq 1} \binom{m}{j} (-1)^j = -1$ to the choice of whether the class meets $\mathcal{J}$, exactly like a single edge. If some edge misses V , then $N_{\mathcal{H}}(V) = 0$: toggling that edge is a sign-reversing involution on the covering subfamilies. Also $N_{\mathcal{H}}(\emptyset) = \mathbf{1}[\mathcal{H} = \emptyset]$. For example, $Q_{\mu}(\{J\}) = \lambda^J - 1$, and for pairwise disjoint edges $Q_{\mu}(\{J_1, \dots, J_m\}) = \prod_i (\lambda^{J_i} - 1)$.

Lemma 3.1 (Cover bound). *For every family $\mathcal{E}$ of cylinder events and every $V \subseteq V_0$,*

$$\|L_V F\|^2 \leq \mathbb{E}_x [N_{\mathcal{H}(x)}(V)^2], \quad \mathcal{H}(x) = \{\text{supp } E : E \in \mathcal{E}, x \in E\}.$$

Consequently $G_F(\lambda) \leq \mathbb{E}_x [Q_{\mu}(\mathcal{H}(x))]$ for all weights $\lambda_v \geq 1$.

Proof. Write $x = (\sigma, y)$ with $\sigma \in \Omega_V$ and $y \in \Omega_{V^c}$, $V^c = V_0 \setminus V$. The operator L_V acts on the V -coordinates only, so $\|L_V F\|^2 = \mathbb{E}_y \|L_V g_y\|_{L^2(\pi_V)}^2$ with $g_y = F(\cdot, y)$. Now g_y is the avoidance indicator on Ω_V of the *restricted* events: those $E \in \mathcal{E}$ whose conditions off V hold at y , restricted to $\text{supp } E \cap V$. Expand

$$g_y = \prod_E (1 - \mathbf{1}_{E|_y}) = \sum_{\mathcal{J}} (-1)^{|\mathcal{J}|} \mathbf{1}_{\mathcal{J}}$$

over subfamilies $\mathcal{J}$ of restricted events, where $\mathbf{1}_{\mathcal{J}}$ is the indicator of their intersection, which is a cylinder on Ω_V with support $\bigcup_{E \in \mathcal{J}} (\text{supp } E \cap V)$, or it is empty. The projection L_V annihilates every function that does not depend on all coordinates in V , hence every cylinder whose support is a proper subset of V . A cylinder with support V is a point indicator $\mathbf{1}_{\{\sigma\}}$. Therefore $L_V g_y = L_V c_y$, where

$$c_y(\sigma) = \sum_{\mathcal{J}: \text{supp } \mathbf{1}_{\mathcal{J}} = V, \mathbf{1}_{\mathcal{J}} = \mathbf{1}_{\{\sigma\}}} (-1)^{|\mathcal{J}|}.$$

Since L_V is an orthogonal projection in $L^2(\pi_V)$, $\|L_V c_y\|^2 \leq \mathbb{E}_\sigma c_y(\sigma)^2$. The subfamilies counted in $c_y(\sigma)$ are exactly the subfamilies of events that hold at $x = (\sigma, y)$ whose traces on V cover V . Two events that hold at x and have the same support are the same cylinder. By the remark on repeated traces, $c_y(\sigma) = N_{\mathcal{H}(x)}(V)$. Taking $\mathbb{E}_y$ gives the first claim. For the second, multiply by μ^V , sum over V and use Lemma 2.2(a). $\square$

The lemma is an equality for $V = \emptyset$ (both sides equal $\mathbb{E}F$). The only loss is the projection step $\|L_V c\| \leq \|c\|$. Two features matter below. First, the suppression is built in: $N_{\mathcal{H}(x)}(V) = 0$ as soon as some event that holds at x misses V . Second, the right-hand side no longer involves the measure, except through the law of $\mathcal{H}(x)$; Theorem 1.1 will follow from a bound on Q_μ that is uniform over all hypergraphs whose edges have weight at most 2.

Remark 3.2 (Unsigned counts are too weak). Lecomte and Tan [7, Fact 9] bound $|\hat{g}(S)|$ for a DNF g on the uniform cube by $2^{|S|} \mathbb{P}_x[S \text{ is covered by the terms satisfied by } x]$; we quote the form from [7], checked earlier. The indicator “ S is covered” does not see cancellations. For example, if $\mathcal{H}(x)$ contains a singleton $\{u\}$ and an edge containing a set $V \not\ni u$, then $N_{\mathcal{H}(x)}(V) = 0$, while $\mathbf{1}[V \text{ is covered}] = 1$, so the bound of [7] charges $2^{|V|}$ at such points.

4. THE HYPERGRAPH INEQUALITY

For a finite multiset $\mathcal{C}$ of subsets of V_0 (empty members allowed) put

$$\Theta_\lambda(\mathcal{C}) = \sum_{\mathcal{J} \subseteq \mathcal{C}} (-1)^{|\mathcal{J}|} \lambda^{\bigcup \mathcal{J}} \quad (\Theta_\lambda(\emptyset) = 1),$$

the sum running over sub-multisets (members are distinguishable).

Lemma 4.1 (Polarization). *Let $\lambda_v \geq 1$, $\mu_v = \lambda_v - 1$, and let $P \subseteq V_0$ be random with $\mathbb{P}(v \in P) = \mu_v/\lambda_v$, independently for different v . For every hypergraph $\mathcal{H}$,*

$$Q_\mu(\mathcal{H}) = \mathbb{E}_P[\Theta_\lambda(\mathcal{H}_P)^2], \quad \mathcal{H}_P = \{J \in \mathcal{H} : J \cap P = \emptyset\}.$$

Proof. Let $\tau(R) = \mathbf{1}[R \cap J \neq \emptyset \text{ for all } J \in \mathcal{H}]$ for $R \subseteq V_0$, and $\hat{\tau}(V) = \sum_{R \subseteq V} (-1)^{|V \setminus R|} \tau(R)$. By inclusion-exclusion,

$$\tau(R) = \sum_{\mathcal{J} \subseteq \mathcal{H}} (-1)^{|\mathcal{J}|} \mathbf{1}[R \cap \bigcup \mathcal{J} = \emptyset]. \quad (2)$$

The Möbius transform of $R \mapsto \mathbf{1}[R \cap U = \emptyset]$ is $V \mapsto (-1)^{|V|} \mathbf{1}[V \subseteq U]$, so $\hat{\tau}(V) = (-1)^{|V|} N_{\mathcal{H}}(V)$. Let

$$\mathcal{T}(p) = \sum_V \hat{\tau}(V) p^V = \sum_R \tau(R) \prod_{v \in R} p_v \prod_{v \notin R} (1 - p_v)$$

be the multilinear extension of τ . If the p_v are independent random variables with mean 0 and variance μ_v , then distinct monomials are orthogonal and $\mathbb{E}[\mathcal{T}(p)^2] = \sum_V \mu^V \hat{\tau}(V)^2 = Q_\mu(\mathcal{H})$, whatever the laws of the p_v . Take $p_v = 1$ with probability μ_v/λ_v and $p_v = -\mu_v$ otherwise. The mean is

$\mu_v/\lambda_v - \mu_v/\lambda_v = 0$ and the variance is $\mu_v/\lambda_v + \mu_v^2/\lambda_v = \mu_v$. Let $P = \{v : p_v = 1\}$. In the second form of $\mathcal{T}$, the factor $1 - p_v$ vanishes for $v \in P$, so only the $R \supseteq P$ contribute; writing $R = P \cup B$ with $B \subseteq P^c$, and using $1 - p_v = \lambda_v$ for $v \notin P$,

$$\mathcal{T}(p) = \sum_{B \subseteq P^c} (-\mu)^B \lambda^{P^c \setminus B} \tau(P \cup B).$$

An edge that meets P meets $P \cup B$, so (2) for the hypergraph $\mathcal{H}_P$ gives $\tau(P \cup B) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \mathbf{1}[B \cap \bigcup \mathcal{J} = \emptyset]$. For $U \subseteq P^c$,

$$\sum_{B \subseteq P^c \setminus U} (-\mu)^B \lambda^{P^c \setminus B} = \lambda^U \prod_{v \in P^c \setminus U} (\lambda_v - \mu_v) = \lambda^U.$$

Hence $\mathcal{T}(p) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \lambda^{\bigcup \mathcal{J}} = \Theta_\lambda(\mathcal{H}_P)$. $\square$

Lemma 4.2 (Matching bound). *If every member of $\mathcal{C}$ has $w_J := \lambda^J \leq 2$, then for every matching $\mathcal{M} \subseteq \mathcal{C}$ (pairwise disjoint members)*

$$|\Theta_\lambda(\mathcal{C})| \leq \prod_{J \in \mathcal{M}} (w_J - 1); \quad \text{in particular } |\Theta_\lambda(\mathcal{C})| \leq 1.$$

Proof. For $v \in V_0$ let $\mathcal{C} - v$ be the multiset of members not containing v , and $\mathcal{C}/v = \{J \setminus \{v\} : J \in \mathcal{C}\}$ (as a multiset). Split the sum over $\mathcal{J}$ according to whether $\mathcal{J}$ contains a member through v . If it does, then $\lambda^{\bigcup \mathcal{J}} = \lambda_v \lambda^{\bigcup (\mathcal{J}/v)}$; the terms without such a member form $\Theta(\mathcal{C} - v)$ both in $\Theta(\mathcal{C})$ and in $\Theta(\mathcal{C}/v)$. Hence

$$\Theta(\mathcal{C}) = \Theta(\mathcal{C} - v) + \lambda_v [\Theta(\mathcal{C}/v) - \Theta(\mathcal{C} - v)] = \lambda_v \Theta(\mathcal{C}/v) - \mu_v \Theta(\mathcal{C} - v). \quad (3)$$

Both families again have all weights at most 2, since $w_{J \setminus v} = w_J/\lambda_v$ for $J \ni v$ and the other members are unchanged. We induct on $|\bigcup \mathcal{C}|$. If $\emptyset \in \mathcal{C}$, then $\Theta(\mathcal{C}) = 0$ (toggling the empty member pairs the terms), and the claim holds. If $\mathcal{C} = \emptyset$, then $\Theta = 1$ and $\mathcal{M} = \emptyset$. This covers the base case $\bigcup \mathcal{C} = \emptyset$. If $\mathcal{M} = \emptyset$ and $\mathcal{C} \neq \emptyset$, it suffices to prove the bound for $\mathcal{M} = \{J\}$ with any $J \in \mathcal{C}$, since $w_J - 1 \leq 1$. So let $J_0 \in \mathcal{M}$ be nonempty, $v \in J_0$, and $\Pi' = \prod_{J \in \mathcal{M} \setminus \{J_0\}} (w_J - 1)$. In $\mathcal{C}/v$ the family $(\mathcal{M} \setminus \{J_0\}) \cup \{J_0 \setminus \{v\}\}$ is a matching, so by induction $|\Theta(\mathcal{C}/v)| \leq (w_{J_0}/\lambda_v - 1)\Pi'$. In $\mathcal{C} - v$ the family $\mathcal{M} \setminus \{J_0\}$ is a matching, so $|\Theta(\mathcal{C} - v)| \leq \Pi'$. By (3), $|\Theta(\mathcal{C})| \leq (w_{J_0} - \lambda_v)\Pi' + \mu_v \Pi' = (w_{J_0} - 1)\Pi'$. $\square$

The hypothesis $w_J \leq 2$ enters only through $w_J - 1 \leq 1$ and its stability under contraction. Without it Θ is unbounded: for m disjoint singletons of weight λ , $\Theta = (1 - \lambda)^m$.

Theorem 4.3 (Hypergraph inequality). *If every edge of a hypergraph $\mathcal{H}$ has $w_J = \lambda^J \leq 2$, then*

$$Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} (w_J - 1) \quad \text{for every matching } \mathcal{M} \subseteq \mathcal{H};$$

in particular $Q_\mu(\mathcal{H}) \leq 1$, and $Q_\mu(\mathcal{H}) \leq \min_{J \in \mathcal{H}} w_J - 1$ if $\mathcal{H} \neq \emptyset$. Equality holds if $\mathcal{H} = \mathcal{M}$ is itself a matching.

Proof. By Lemma 4.1, and Lemma 4.2 applied to $\mathcal{H}_P$ with the matching $\mathcal{M} \cap \mathcal{H}_P$, $Q_\mu(\mathcal{H}) \leq \mathbb{E}_P \prod_{J \in \mathcal{M} \cap \mathcal{H}_P} (w_J - 1)^2$. The members of $\mathcal{M}$ are disjoint, so the events $\{J \cap P = \emptyset\}$, $J \in \mathcal{M}$, are independent, with probabilities $\prod_{v \in J} (1 - \mu_v/\lambda_v) = 1/w_J$. Hence

$$Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} \left[\left(1 - \frac{1}{w_J}\right) + \frac{(w_J - 1)^2}{w_J} \right] = \prod_{J \in \mathcal{M}} (w_J - 1).$$

The equality case is the example after the definition of Q_μ . $\square$

Proof of Theorem 1.1. By Lemma 3.1, $G_F(\lambda) \leq \mathbb{E}_x[Q_\mu(\mathcal{H}(x))]$. Every edge of $\mathcal{H}(x)$ is the support of an event, so it has weight at most 2, and Theorem 4.3 gives $Q_\mu(\mathcal{H}(x)) \leq 1$ for every x . $\square$

Theorem 4.3 with the matching bound gives slightly more: $G_F(\lambda) \leq \mathbb{E}_x \min_{\mathcal{M}} \prod_{J \in \mathcal{M}} (w_J - 1)$, the minimum over matchings in $\mathcal{H}(x)$. It would be interesting to know whether the minimum over matchings can be replaced by a fractional-matching version (Conjecture 6.5).

5. ENERGY TAILS AND DNFs

Proof of Corollary 1.2. Take $\lambda_v = \lambda = 2^{1/k}$ for all v ; then $w_E = \lambda^{|\text{supp } E|} \leq 2$, and Theorem 1.1 gives $G_F(\lambda) \leq 1$. Let $a = \mathbb{P}(F = 0) = \mathbb{E}h$. Since $\|F^{\emptyset}\|^2 = (\mathbb{E}F)^2 = (1 - a)^2$,

$$\sum_{U \neq \emptyset} \lambda^{|U|} \|F^{\neg U}\|^2 \leq 1 - (1 - a)^2 = a(2 - a).$$

For $t \geq 0$ every U with $|U| > t$ is nonempty and has $\lambda^{|U|} \geq 2^{(t+1)/k}$, so $\text{En}(F; t) \leq 2^{-(t+1)/k} a(2 - a)$. Finally $(1 - F)^{\neg U} = -F^{\neg U}$ for $U \neq \emptyset$. $\square$

Proof of Corollary 1.3. Let g be computed by the DNF $T_1 \vee \dots \vee T_m$ with $-1 = \text{True}$, and let E_i be the set of inputs that satisfy the term T_i . A term is a conjunction of at most k literals on distinct variables (a term containing x_v and $\bar{x}_v$ is never satisfied and may be dropped), so E_i is a cylinder event with $|\text{supp } E_i| \leq k$. Then $g = 2F - 1$ with $F = \prod_i (1 - \mathbf{1}_{E_i})$, and $p = \mathbb{P}[g = -1] = \mathbb{P}(F = 0)$. As $g^{\neg U} = 2F^{\neg U}$ for $U \neq \emptyset$, the first claim is Corollary 1.2 multiplied by 4, and $p(2 - p) \leq 1$. If $4 \cdot 2^{-(d+1)/k} \leq \varepsilon$, i.e. $d + 1 \geq k \log_2(4/\varepsilon)$, then $W^{>d}[g] \leq \varepsilon$; the least such integer d is at most $k \log_2(4/\varepsilon)$.

For the weighted sum, $\|g^{\emptyset}\|^2 = (1 - 2p)^2$ and, with $\lambda = 2^{1/k}$,

$$\sum_{U \neq \emptyset} \lambda^{|U|} \|g^{\neg U}\|^2 = 4(G_h - p^2),$$

where $G_h = G_F - 1 + 2p \leq 2p$ by Lemma 2.2(c). So the sum is at most $(1 - 2p)^2 + 8p - 4p^2 = 1 + 4p$. For the influence, $\sum_U (\lambda^{|U|} - 1) \|F^{\neg U}\|^2 = G_F - \mathbb{E}F \leq p$, and $\lambda^{|U|} - 1 \geq |U| \ln \lambda = |U| \ln 2/k$. Hence

$$\mathbf{I}[g] = \sum_U |U| \|g^{\neg U}\|^2 = 4 \sum_U |U| \|F^{\neg U}\|^2 \leq (4k/\ln 2) p.$$

For a q -ary DNF, a term $\bigwedge_{v \in S} (x_v \in S_v)$ is the disjoint union of the cylinder events $\{x_S = \sigma\}$, $\sigma \in \prod_{v \in S} S_v$, which all have support S . Replacing each term by these events changes neither F nor the supports. $\square$

For a width- k CNF h , $-h$ is a width- k DNF and $W^{>t}[h] = W^{>t}[-h]$; so the same bounds hold with $p = \mathbb{P}[h = +1]$ (the probability that h is false), and the bound $4 \cdot 2^{-(t+1)/k}$ holds as stated. Here $\mathbf{I}[g] = \sum_U |U| \|g^{=U}\|^2 = \sum_v \|L_v g\|^2$ is the Efron–Stein total influence; it is the usual total influence for the uniform measure.

Remark 5.1 (The influence bound is elementary). Let g be a width- k DNF and $p = \mathbb{P}[g = -1]$. If $g(x) = -1$, fix a term satisfied at x ; changing a coordinate outside it leaves $g = -1$. So the fibre of x in direction v is non-constant for at most k coordinates v . Since $\|L_v g\|^2 = \mathbb{E} \text{Var}_{x_v}(g)$ and $\text{Var}_{x_v}(g) = 4q(1-q) \leq 4q$ on a non-constant fibre, with q the conditional probability of $g = -1$, we get $\sum_v \|L_v g\|^2 \leq 4kp$ under any product measure. On the uniform cube, $\mathbf{I}[g]$ is the mean number of sensitive coordinates; every sensitive edge has exactly one endpoint with $g = -1$, so $\mathbf{I}[g] \leq 2kp$. Corollary 1.3 gives the weaker $(4k/\ln 2)p$.

Remark 5.2 (Non-uniform widths). Theorem 1.1 also handles terms of very different widths, and weights adapted to the variables. For instance, if every term contains at most k_1 variables from a set A and at most k_2 from its complement, then $\lambda_v = 2^{1/(2k_1)}$ on A and $2^{1/(2k_2)}$ off A are admissible. This gives $\sum \|F^{=U}\|^2 \leq 2^{-\tau}$ over the U with $|U \cap A|/k_1 + |U \setminus A|/k_2 > 2\tau$. The weights used in Section 8 are of this kind, with $\log \lambda_v$ proportional to the logarithm of the size of the alphabet at v .

Remark 5.3 (What is not claimed). Corollary 1.3 concerns concentration on low *degree* only. It says nothing about the number of significant coefficients (Mansour’s sparsity bound and Mansour’s conjecture [9], [7]), and it does not give the random-restriction (decision-tree depth) form of the switching lemma. It does not apply to depth-3 circuits or to AC^0 ; we do not know whether the method extends to them.

6. SHARPNESS, LIMITS OF THE METHOD, OPEN PROBLEMS

Proposition 6.1 (The base $2^{-1/k}$ is sharp). *Fix $k \geq 1$ and $j \geq 1$, and let $t = jk - 1$. For every $\eta > 0$ there is a family of cylinder events of width k , on a product space with all alphabets of size 2 and a suitable product measure, or on $[q]^n$ with the uniform measure and q large, such that*

$$\text{En}(F; t) \geq \frac{2^{-(t+1)/k}}{e\sqrt{j}} - \eta.$$

Consequently, no bound $\text{En}(F; t) \leq C\rho^{-(t+1)/k}$ with $\rho > 2$ holds for all width- k systems, and the same is true for $W^{>t}[g]$ of width- k DNFs under biased product measures on $\{\pm 1\}^n$.

Proof. Take m events on pairwise disjoint sets $S_1, \dots, S_m$ of k coordinates, each fixing at every coordinate of its support a value of probability p (the value -1 of a p -biased bit, or one value of $[q]$ with $q = 1/p$), and put $\pi = p^k$. The factors $1 - \mathbf{1}_{E_i}$ are independent, so $F^{=U} = \prod_i (1 - \mathbf{1}_{E_i})^{=U \cap S_i}$. If U is the union of j of the S_i , Lemma 2.2(d) gives $\|F^{=U}\|^2 = [\pi(1-p)^k]^j (1-\pi)^{2(m-j)}$, and $|U| = jk$. Hence

$$\text{En}(F; jk - 1) \geq \binom{m}{j} [\pi(1-p)^k]^j (1-\pi)^{2(m-j)} \rightarrow e^{-2s} \frac{s^j}{j!}$$

as $p \rightarrow 0$ with $m = \lfloor s/\pi \rfloor$. With $s = j/2$ and $j! \leq e j^{j+1/2} e^{-j}$ the limit is at least $2^{-j}/(e\sqrt{j})$, and $2^{-j} = 2^{-(t+1)/k}$. $\square$

So at most a factor of order $\sqrt{t/k}$ could be gained in Corollary 1.2 over general product spaces. Numerically, the supremum over m of $\text{En}(F; jk - 1) 2^j$ in this family is about $2/\sqrt{2\pi j}$ for small p (an earlier internal computation; we do not prove it). Together with Proposition 2.3 this shows that both the threshold 2 in Theorem 1.1 and the base of Corollary 1.2 are optimal at the level of generality of the theorem.

Example 6.2 (The uniform cube). On $\{\pm 1\}^n$ with the uniform measure the extremal families above are not available ($p = 1/2$), and the truth may be better. Let g be the OR of s parities on pairwise disjoint blocks $B_1, \dots, B_s$ of k variables. Each parity is a width- k DNF (with 2^{k-1} terms), so g is a width- k DNF. If the i -th parity is “true” when $\chi_{B_i} = -1$, then $F = \prod_i (1 + \chi_{B_i})/2$, and $\text{En}(F; jk - 1) = 4^{-s} \sum_{i \geq j} \binom{s}{i}$. Taking s near $4j/3$ and using Stirling’s formula, $\max_s 4^{-s} \binom{s}{j} = 3^{-j+o(j)}$. So the exponential rate of the best width- k bound on the uniform cube lies between $3^{-t/k}$ and $2^{-t/k}$. We do not know where. In random and hill-climbing searches over small Boolean systems (earlier internal computations) we found nothing better than this example.

No one-event-at-a-time proof. It is natural to try to prove Theorem 1.1 by adding the events one at a time. The following example shows that this cannot work in the obvious form. Adding a further event of weight at most 2 to a system can *increase* G_F .

Example 6.3 (Adding an event can increase G_F). Let $n = 2$, $\Omega = [8]^2$ with the uniform measure, $\lambda_1 = \lambda_2 = \sqrt{2}$, and let $\mathcal{E}$ consist of the 49 point events $\{x = (a, b)\}$ with $a \neq 0 \neq b$. Let $A = \{x = (0, 0)\}$. All events have weight 2, and $G_{F_{\mathcal{E}}} < G_{F_{\mathcal{E} \cup \{A\}}}$; numerically the values are 0.31571 and 0.31757. (The exact values are computed in Section 9.) A similar example exists on $[5]^3$, using the full-width cylinders with an even number ≥ 2 of coordinates different from 0.

Also the inequality fails beyond the threshold. For n uniform bits, the OR of the n literals $x_v = -1$ and $\lambda_v = 4$ give $G_F = (5/4)^n$ by Lemma 2.2(d) and Proposition 2.3. We state two strengthenings that are supported by numerical evidence but are open.

Conjecture 6.4 (Excess weights). *For every family of cylinder events and all weights $\lambda_v \geq 1$,*

$$G_F(\lambda) \leq \prod_{E \in \mathcal{E}} (1 + \mathbb{P}(E)(w_E - 2)_+).$$

Conjecture 6.5 (Fractional matchings). *If every edge of $\mathcal{H}$ has $w_J \leq 2$, then $Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{H}} (w_J - 1)^{y_J}$ for every fractional matching y ($y_J \geq 0$, $\sum_{J \ni v} y_J \leq 1$ for all v).*

Both were tested numerically on small random systems (several thousand cases on $[q]^n$ with $q \leq 5$, $n \leq 9$; floating-point internal computations, with a numerical linear program over y for Conjecture 6.5; not part of this note), without a violation. For Conjecture 6.4 only the weaker form $G_F \leq \exp(\sum_E \mathbb{P}(E)(w_E - 2)_+)$ was tested. For Conjecture 6.5 equality (up to rounding) is frequent. Example 6.3 shows that a proof of Conjecture 6.4 by induction on the number of events would need a stronger induction hypothesis.

Open problems. (1) Determine the optimal rate for width- k DNFs on the uniform cube (Example 6.2). (2) Is there an analogue of Theorem 1.1 for depth-3 circuits, or for AC^0 , with a weight condition on the bottom layer? (3) Does the signed cover count give sparsity bounds in the direction of Mansour's conjecture?

7. WEIGHTS ALONG A FILTRATION

In Theorem 1.1 the cost of a component $F^{=U}$ is a product over the coordinates in U . In the application the natural cost of a function of a residue $x_\ell \bmod \ell^f$ depends on *how many* base- ℓ digits it reads, and an event that fixes $x_\ell \bmod \ell^v$ fixes the first v digits. We now prove the corresponding variant.

Setting. Let $\mathcal{P}$ be a finite index set. Each coordinate X_ℓ ($\ell \in \mathcal{P}$) is a tuple of independent *digits* $(\delta_{\ell,i})_{i_0(\ell) \leq i < f_\ell}$ with arbitrary distributions on finite sets, and the coordinates are independent. The Efron–Stein decomposition $F = \sum_U F^{=U}$ is taken over the digits, so U is a set of pairs (ℓ, i) ; we write $U_\ell = \{i : (\ell, i) \in U\}$. An event E is a *digit-prefix event* if, for some integers $0 \leq v_\ell(E) \leq f_\ell$, it fixes exactly the digits $\delta_{\ell,i}$ with $i_0(\ell) \leq i < v_\ell(E)$, for each ℓ , to arbitrary values (no digit of X_ℓ if $v_\ell(E) \leq i_0(\ell)$). The offsets are integers $i_0(\ell) \geq 0$. Such an event is a cylinder event on the digits. (The offset $i_0(\ell)$ allows digits below $i_0(\ell)$ to be fixed in advance; they are not coordinates.)

Theorem 7.1 (Filtration energy bound). *Let $\lambda_\ell \geq 1$ ($\ell \in \mathcal{P}$), and suppose that every event of $\mathcal{E}$ is a digit-prefix event with $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$. Then*

$$\sum_U \left(\prod_{\ell: U_\ell \neq \emptyset} \lambda_\ell^{1 + \max U_\ell} \right) \|F^{=U}\|^2 \leq 1.$$

Applied directly, Theorem 1.1 with digit weights would charge a set U for each digit in it, and the event weights would grow with the number of fixed digits. Here the charge for U_ℓ depends only on its *deepest* digit, which is what a function of X_ℓ that reads U_ℓ costs, and the hypothesis loses only the factor 2 in the exponent.

Proof. Write $\Lambda_{\ell,i} = \lambda_\ell^{i+1}$ for $i \geq i_0(\ell)$, $\Lambda_{\ell,i_0(\ell)-1} = 1$, and $\mu'_{\ell,j} = \Lambda_{\ell,j} - \Lambda_{\ell,j-1} \geq 0$ for $i_0(\ell) \leq j < f_\ell$. For such j let $P_{\ell,j} = I - \mathbb{E}_{B_{\ell,j}}$, where $\mathbb{E}_{B_{\ell,j}}$ averages over the *tail block* $B_{\ell,j} = (\delta_{\ell,i})_{i \geq j}$. Averaging over a digit in U kills $F^{=U}$, and $F^{=U}$ does not depend on the digits outside U . Hence $P_{\ell,j}F^{=U} = F^{=U}$ if $U_\ell \neq \emptyset$ and $\max U_\ell \geq j$, and $P_{\ell,j}F^{=U} = 0$ otherwise. Call a set V of pairs (ℓ, j) a *selection* if it contains at most one pair for each ℓ , and put $L'_V = \prod_{(\ell,j) \in V} P_{\ell,j}$. Then $\|L'_V F\|^2 = \sum_{U \succ V} \|F^{=U}\|^2$, where $U \succ V$ means $\max U_\ell \geq j$ for all $(\ell, j) \in V$. For fixed U , the sum of μ'^V over the selections V with $U \succ V$ factorises and telescopes:

$$\prod_{\ell} \left(1 + \sum_{i_0(\ell) \leq j \leq \max U_\ell} \mu'_{\ell,j}\right) = \prod_{\ell: U_\ell \neq \emptyset} \Lambda_{\ell, \max U_\ell}.$$

So the left side of the claim equals $\sum_V \mu'^V \|L'_V F\|^2$, the sum over selections.

Fix a selection V and pass to the coarser product space whose coordinates are, for $(\ell, j) \in V$, the head $(\delta_{\ell,i})_{i < j}$ and the tail block $B_{\ell,j}$, and, for the other ℓ , the whole X_ℓ . There L'_V is the operator $L_{V'}$ of Section 2 for the set V' of tail blocks. Each event E is a disjoint union of cylinder events on the coarse coordinates with the same coarse support: enumerate the values of the unfixed digits of each coarse coordinate that E involves. Splitting every event in this way does not change F , and exactly one piece of E holds at x if E does. The trace of the coarse support of E on V' is the set of the $B_{\ell,j}$ with $(\ell, j) \in V$ and $j < v_\ell(E)$. Identify $B_{\ell,j}$ with the pair (ℓ, j) and put $\hat{E} = \{(\ell, j) : i_0(\ell) \leq j < v_\ell(E)\}$; then this trace is $\hat{E} \cap V$. Since N depends only on traces, Lemma 3.1 gives

$$\|L'_V F\|^2 \leq \mathbb{E}_x [N_{\hat{\mathcal{H}}(x)}(V)^2], \quad \hat{\mathcal{H}}(x) = \{\hat{E} : E \in \mathcal{E}, x \in E\}.$$

Sum over selections and add the nonnegative terms of all other sets V of pairs: the left side of the claim is at most $\mathbb{E}_x [Q_{\mu'}(\hat{\mathcal{H}}(x))]$. By Theorem 4.3 it remains to show that $\prod_{(\ell,j) \in \hat{E}} (1 + \mu'_{\ell,j}) \leq 2$ for every event. Fix ℓ , write $\lambda = \lambda_\ell$, $v = v_\ell(E)$, $i_0 = i_0(\ell)$, and assume $v > i_0$. Here $1 + \mu'_{\ell,i_0} = \lambda^{i_0+1}$, and $1 + \mu'_{\ell,j} = 1 + \lambda^j(\lambda - 1)$ for $j > i_0$ (if $v \leq i_0$ the factor of ℓ is 1). The hypothesis gives $\lambda^{2v} \leq 2$, so $\lambda^j \leq \lambda^v \leq \sqrt{2} < \lambda + 1$ for $j < v$, which is equivalent to $1 + \lambda^j(\lambda - 1) \leq \lambda^2$. The factor of ℓ is therefore at most $\lambda^{i_0+1} \lambda^{2(v-i_0)} \leq \lambda^{2v}$, and the product over ℓ is at most $\prod_{\ell} \lambda_\ell^{2v_\ell(E)} \leq 2$. $\square$

Corollary 7.2 (Modulus-weighted tail). *In Theorem 7.1 let the $\ell \in \mathcal{P}$ be primes, let $T > 1$, $\mathcal{L} = \log T$, and suppose that $\prod_{\ell} \ell^{v_\ell(E)} \leq T$ for every event. For a set U of digits put $m_U = \prod_{\ell: U_\ell \neq \emptyset} \ell^{1+\max U_\ell}$. Then for every*

$$\tau \geq 0$$

$$\sum_{U: m_U > e^\tau} \|F^{=U}\|^2 \leq 2^{-\tau/(2\mathcal{L})}.$$

Proof. Take $\lambda_\ell = 2^{\log \ell/(2\mathcal{L})}$. Then $\prod_\ell \lambda_\ell^{2v_\ell(E)} = 2^{\log \prod_\ell \ell^{v_\ell(E)}/\mathcal{L}} \leq 2$, and the weight of U in Theorem 7.1 is $2^{\log m_U/(2\mathcal{L})}$; apply Lemma 2.2(b). $\square$

Remark 7.3 (The factor 2 in the exponent). The proof gives more. If $v > i_0$, the factor of ℓ is

$$\lambda^{i_0+1} \prod_{i_0 < j < v} (1 + \lambda^j(\lambda - 1)) \leq \exp\left(\lambda^{i_0+1} - 1 + \sum_{i_0 < j < v} \lambda^j(\lambda - 1)\right) = \exp(\lambda^v - 1).$$

So it suffices that $\sum_\ell (\lambda_\ell^{v_\ell(E)} - 1) \leq \ln 2$ for every event. In Corollary 7.2 take $\lambda_\ell = 2^{\log \ell/(\rho\mathcal{L})}$ and put $x_\ell = v_\ell(E) \log \ell/\mathcal{L}$, so that $\sum_\ell x_\ell \leq 1$. The function $x \mapsto 2^{x/\rho} - 1$ is convex and vanishes at 0, so it is superadditive, and $\sum_\ell (2^{x_\ell/\rho} - 1) \leq 2^{1/\rho} - 1$. This is at most $\ln 2$ when $\rho \geq \rho_0 := 1/\log_2(1 + \ln 2) = 1.3163\dots$, and then the tail in Corollary 7.2 is at most $2^{-\tau/(\rho\mathcal{L})}$.

8. APPLICATION: ERDŐS–STRAUS WITNESS MODULI

This section explains where Theorem 7.1 is used; the details are in [12], and nothing here is needed for the rest of the note. The Erdős–Straus conjecture asserts that $4/n = 1/x + 1/y + 1/z$ has a solution in positive integers for every $n \geq 2$; it is open. For $M \equiv 3 \pmod{4}$ the classical multiplier identity gives a solution for every n with $n \equiv -uv^{-1} \pmod{M}$ for some factorisation $uvw = (M+1)/4$. Let $W(n)$ be the least such M ($+\infty$ if there is none). In [12] it is shown, assuming Gallagher’s prime number theorem for the family of all characters and four estimates of Elsholtz and Tao on divisor sums, that $W(p) \geq \exp(c(\log p)^{1/5}(\log \log p)^{-1/5})$ for infinitely many primes p . This is an Ω -result about one particular family of explicit solutions, and says nothing about whether the conjecture holds.

The structure of the argument is as follows. Fix T and $\mathcal{L} = \log T$. After a *quarantine* $n \equiv 1 \pmod{Q}$, the condition $W(n) > T$ holds as soon as n avoids a finite family of congruence events $n \equiv b_i \pmod{r_i}$ with $r_i \leq T$, at most T^2 of them. For n distributed by Haar measure on the unit group $\widehat{\mathbb{Z}}^\times$, conditioned on the quarantine, the residues $X_\ell = n \bmod \ell^{f_\ell}$ at the primes involved are independent. Each X_ℓ is uniform on the units modulo ℓ^{f_ℓ} , or on a fibre $1 + \ell^a \mathbb{Z}$ modulo ℓ^{f_ℓ} if ℓ is quarantined with $a = a_\ell \geq 1$. On these sets the uniform measure is the product of the uniform distributions of the free base- ℓ digits (digit 0 ranging over the nonzero residues in the unit case), so the digits are independent coordinates. Moreover an event $n \equiv b \pmod{r}$ is a digit-prefix event with $\prod_\ell \ell^{v_\ell(E)} = r \leq T$. So the avoidance indicator F is in the setting of Corollary 7.2, and so is each conditioned function $F^{(j)}$ (the indicator that no earlier event occurs, restricted to the cylinder of the j -th event): there the digit offsets $i_0(\ell)$ are raised.

To find a *prime* p with $W(p) > T$ one needs a minorant $B \leq F$ that is a combination of congruence classes of small modulus, has mean comparable to $\mathbb{E}F$, and has small ℓ^1 norm. [12] uses the one-sided ℓ^2 sandwich of Bazzi [2] in the form of Razborov [11]:

$$B = 1 - \sum_i \mathbf{1}_{E_i} \left(1 - \sum_{j < i} \mathbf{1}_{E_j} u_j \right)^2 \leq F,$$

$$\mathbb{E}[F - B] \leq m^2 \sum_j \mathbb{P}(E_j) \mathbb{E}[(F^{(j)} - u_j)^2],$$

for any functions u_j of the coordinates not fixed by E_j . The natural choice of u_j is the truncation of the Efron–Stein decomposition of $F^{(j)}$ to digit sets U with $m_U \leq e^\tau$. By (1) it is a combination of classes of modulus at most e^τ , and by Corollary 7.2 its error is at most $2^{-\tau/(2\mathcal{L})}$. The sandwich needs the error to be about $e^{-O(S)}m^{-2}$, where S is the total mass of the events. So $\tau \ll \mathcal{L}(S + \mathcal{L})$ suffices; in [12] $S \ll \mathcal{L}^4$ (using the Elsholtz–Tao input), and the junta moduli satisfy $\log d_i \ll \mathcal{L}^5$. The quarantine costs $\log Q \ll \mathcal{L}^5 \log \mathcal{L}$, and a linear transfer theorem based on Gallagher’s theorem then produces a prime p with $\log p \ll \mathcal{L}^5 \log \mathcal{L}$.

An earlier version of [12] encoded the residues in binary and used the switching-lemma tail of LMN instead. There the junta modulus picked up factors for the width of the events and for the bit length, and it was one of the two terms of the largest order (the other being the quarantine). With the energy bound the junta term is smaller than the quarantine term. The exponent $1/5$ needs two further ingredients of [12] (a graded quarantine and a moment bound), and we do not attribute it to Theorem 7.1 alone. Two features of the energy bound matter here: the events are congruences modulo integers up to T , so the alphabets are huge and the measure is far from the Boolean setting, and the cost of a junta must be measured by its modulus rather than by the number of primes it involves.

9. COMPUTATIONAL CHECKS

The proofs above are complete. The script `scripts/energy_note_check.py` is a sanity check of the statements on small cases. It uses exact rational arithmetic throughout; where a statement involves an irrational number, the comparison is rewritten as an exact rational one (for instance $W^{>t} \leq c 2^{-(t+1)/k}$ as $(W^{>t})^k 2^{t+1} \leq c^k$). With seed 1 it checks:

- (A) Theorem 1.1, Lemma 2.2(a) and Lemma 3.1 (for every V) on 150 random systems on $\prod_v [q_v]$ ($n \leq 4$ coordinates, $q_v \leq 5$), with random rational product measures and random rational weights in $[1, 2]$;
- (B) the identity of Lemma 4.1, the two formulas for $N_{\mathcal{H}}$, and the matching bounds of Lemma 4.2 and Theorem 4.3 (including the equality case) on 300 random weighted hypergraphs on at most 5 vertices;
- (C) Corollary 1.3 for *every* Boolean function on 4 bits under the uniform measure, with k its minimal DNF width: the tail bound for

- all t (exactly), the bounds $1 + 4p$ and $(4k/\ln 2)p$ at the endpoint $\lambda = 2^{1/k}$ (exactly for $k = 1$, in floating point with slack 10^{-12} otherwise), and $G_F(\lambda) \leq 1$ at a rational $\lambda \leq 2^{1/k}$; the largest value of $W^{>t}[g] 2^{(t+1)/k} / (4p(2-p))$ is 0.680;
- (D) the same statements on 150 random DNFs under biased product measures on $\{\pm 1\}^n$ and q -ary DNFs with set-valued literals;
 - (E) Theorem 7.1 on 150 random digit-prefix systems in bases 2 and 3, with random offsets $i_0(\ell) \in \{0, 1\}$ and random digit distributions;
 - (F) the level polynomial of Proposition 6.1 against brute force, Corollary 1.2 along that family, the formula of Example 6.2, and the exact sign in Example 6.3. The values $(\max_s \text{En}(F; jk - 1))^{1/j}$ of Example 6.2 (0.299, 0.311, 0.319 for $j = 10, 20, 40$) are printed, not checked.

The run takes about six minutes on one core; its output is in the file `data/energy_note/check_seed1.txt`.

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